

陈亦新

北京大学博士研究生

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研究方向

我研究医学影像分割与 PET、CT、MRI 之间的跨模态学习，也开展肿瘤、脂肪肝和多器官代谢的临床影像研究。

教育经历

北京大学 2023 年 9 月至今
医学技术博士研究生

伦敦国王学院 2019–2020 年
人工智能硕士，Distinction

厦门大学 2015–2019
自动化本科，数理金融第二学位

挑战赛成绩与科研项目

autoPET V: Category 1 第二名 2026
MICCAI autoPET V 挑战赛，交互式 PET/CT 病灶分割。

CMR-MULTI: 第二名 2026
多序列心脏 MRI 分割，方法论文入选 CMRSeg 挑战赛口头报告。

REG²: 第四名 2026
方法论文入选 MICCAI 2026 挑战赛 workshop 海报。

国家自然科学基金博士生项目 2026–2027，青年学生基础研究项目（博士研究生）
项目负责人，研究全人体 PET/CT 的多器官代谢信息提取与分析。

工作经历

鹰瞳科技 AirDoc 2022 年 7 月–9 月
注册算法工程师

研发 ICL 术后拱高预测模型，利用眼部测量与晶体尺寸，预测手术植入晶体与自身晶状体之间的距离。
开展超广角眼底图像分析、临床需求调研，并参与医疗器械注册工作。

安德医智 BioMind 2020 年 11 月–2022 年 6 月
医疗图像深度学习算法工程师

开发心脏超声模型，负责图像质控、多普勒频谱分割与测量、间隔缺损检测及瓣膜反流分析。
与医生梳理诊断流程，将模型接入超声辅助诊断系统，并参与卵巢肿块智能诊疗系统研发。

2026

Whole-Body [¹⁸F]FDG PET/CT Characterizes Multi-Organ Metabolic Alterations in MASLD

EJNMMI, 2026
已接收

Yixin Chen (第一作者) , Chuanzhi Zhou, Lijuan Guo, et al.

NeuronCtrl: Geometry-Aware Safe Closed-Loop Generative Control for Neuronal Microenvironment Dynamics

ICML, 2026
Spotlight

Haowei Xu, Yixin Chen, W. Fu, Hongbin Han, Zhaoheng Xie
会议 OpenReview

Neural Network Approximation Aided Kinetic Parameter Inference of Compartment Models in Dynamic PET

IEEE TRPMS, 2026

Wenrui Shao, Yixin Chen, Ourui Fu, Tian Li, Zhi Yang, Nan Li, Hongbin Han, Xiangxi Meng, Zhaoheng Xie
论文

PHATE-Net: Differentiable Pseudotime Learning for Trustworthy Disease Trajectories in PET

CVPR Findings, 2026

Yixin Chen, Yan Wang, Wenrui Shao, Zhaoheng Xie
论文

2025

Modality-projection universal model for comprehensive full-body medical imaging segmentation

Nature
Communications, 2025

Yixin Chen (第一作者) , Lin Gao, Yajuan Gao, Rui Wang, Jingge Lian, Xiangxi Meng, Yanhua Duan, Leiying Chai, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie
论文 arXiv 代码

VP-SFDA: Visual Prompt Source-Free Domain Adaptation for Cross-Modal Medical Image

Health Data Science,
2025

Yixin Chen, Yan Wang, Zhaoheng Xie
论文

Neural network-aided unsupervised input function estimation for dual-time-window PET Patlak analysis

EJNMMI Physics, 2025

Wenrui Shao, Yarong Zhang, Fen Du, Fangxiao Cheng, Yixin Chen, Xiangxi Meng, Ying Liang, Zhaoheng Xie
论文

A Physiology-Supervised Neural Operator Framework for Kinetic Parameter Estimation in Dynamic PET

IEEE NSS/MIC/RTSD,
2025

W. Shao, Yixin Chen, Z. Wang, R. Xu, X. Meng, Z. Xie
论文

2024

IEEE TMI, 2024

PCNet: Prior Category Network for CT Universal Segmentation Model

Yixin Chen, Yajuan Gao, Lei Zhu, Wenrui Shao, Yanye Lu, Hongbin Han, Zhaoheng Xie
论文 代码

MICCAI, 2024
海报报告

LUCIDA: Low-Dose Universal-Tissue CT Image Domain Adaptation for Medical Segmentation

Yixin Chen, Xiangxi Meng, Yan Wang, Shuang Zeng, Xi Liu, Zhaoheng Xie
论文 代码

IEEE TRPMS, 2024

Structure-Enhanced Unsupervised Domain Adaptation for CT Whole-Brain Segmentation

Yixin Chen (第一作者), Yajun Gao, Lei Zhu, Jianan Li, Yan Wang, Jiakui Hu, Hongbin Han, Yanye Lu, Zhaoheng Xie
论文 代码

IEEE NSS/MIC/RTSD,
2024

Neural Network-Assisted Dual-time-window Acquisition: Modeling Incomplete Input Functions for Patlak Analysis

W. Shao, Y. Zhang, **Yixin Chen**, X. Meng, Y. Liang, Z. Xie
论文

2023

IEEE TMI, 2023

FVP: Fourier Visual Prompting for Source-Free Unsupervised Domain Adaptation of Medical Image Segmentation

Yan Wang, Jian Cheng, **Yixin Chen**, Shuai Shao, Lanyun Zhu, Zhenzhou Wu, Tao Liu, Haogang Zhu
论文

IEEE EMBC, 2023

Rethinking Disentanglement in Unsupervised Domain Adaptation for Medical Image Segmentation

Yan Wang, **Yixin Chen** (第一作者), Yingying Zhang, Haogang Zhu
论文

Frontiers in
Cardiovascular
Medicine, 2023

Echocardiography-based AI for detection and quantification of atrial septal defect

Xixiang Lin, Feifei Yang, **Yixin Chen** (第一作者), et al.
论文

2022

IEEE EMBC, 2022

MSGAN: Multi-Stage Generative Adversarial Networks for Cross-Modality Domain Adaptation

Yan Wang, **Yixin Chen** (第一作者), Wenjun Wang, Haogang Zhu
论文

Cancers, 2022

A deep learning model system for diagnosis and management of adnexal masses

Jingbo Li, **Yixin Chen** (第一作者), Minyu Zhang, et al.
论文

Echocardiography-based AI detection of regional wall motion abnormalities and quantification of cardiac function in myocardial infarction		Frontiers in Cardiovascular Medicine, 2022
Xixiang Lin, Feifei Yang, Yixin Chen , et al.		
论文		
预印本		
		arXiv, 2026
PreSight: Preoperative Outcome Prediction for Parkinson's Disease via Region-Prior Morphometry and Patient-Specific Weighting		
Yand Wang, Chen Zhang, Lanyun Zhu, Yixin Chen , Qunbo Wang, Yutong Bai, Jurgen Germann, Yinghong Wen, Shuai Shao		
arXiv		
		SSRN 预印本, 2025
Whole-Body Inter-Organ Metabolic Atlas for PET/CT Anomaly Detection		
Yixin Chen , Lin Gao, Lei Zhu, Ruoyan Xu, Fangxiao Cheng, Xinyu Liu, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie		
预印本		
		arXiv 预印本, 2023
FP-PET: Large Model, Multiple Loss And Focused Practice		
Yixin Chen , Ourui Fu, Wenrui Shao, Zhaoheng Xie		
arXiv		
		arXiv, 2023
SL: Stable learning in source-free domain adaption for medical image segmentation		
Yixin Chen , Yan Wang		
arXiv		
挑战赛论文与会议摘要		
Diagnosis-Conditioned Retrieval for Structured Pathology Workflows		MICCAI REG ² challenge workshop, 2026 海报报告
Yixin Chen		
论文		
Acquisition-Structured nnU-Net Ensembles for Multi-Sequence Cardiac MRI Segmentation and Quantification		MICCAI CMRSeg workshop, 2026 入选口头报告
Yixin Chen , Yizhen Peng		
论文		
Failure-Mode-Driven Multimodal Mitral Valve Segmentation for the MVAA 2026 Challenge		MICCAI MVAA challenge workshop, 2026 海报报告
Yixin Chen		
论文		
Multiorgan Metabolic Dysregulation in Alzheimer's Disease: Total-body PET/CT Reveals Systemic Network Interactions with Amyloid Burden		EANM, 2025 口头报告摘要, OP-308
R. Xu, Y. Duan, L. Gao, Yixin Chen , H. Han, Z. Cheng, Z. Xie		
会议日程		

专利、教学与学术服务

12 项发明专利

4 项已授权，8 项实质审查中。

Nature Communications

Early Career Co-Reviewer, 合作审稿

计算机视觉课程讲师

录制初级与中级课程，讲解机器学习基础及视觉算法。

《深度学习理论与实战：PyTorch 案例详解》

2021

独著，清华大学出版社，2021 年。